Reproducibility in Practice

Week 9: CS 203 - Software Tools and Techniques for AI

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The Friday Night Nightmare

11:30 PM — You (on WhatsApp):

> *"Done! Model hit 94% accuracy. Just pushed* `model_final_v2.py`*. We are going to ace this."*

11:45 PM — Your partner:

> *"I pulled it. I'm only getting 81%."*

11:47 PM — You:

> *"Wait, let me run it again... I'm getting 86% now?! What hyperparameters did I use an hour ago?"*

11:55 PM — Your partner:

> *"I tried to fix it. Now I get* `ModuleNotFoundError: No module named 'xgboost'`*. I'm on a Mac, are you on Windows?"*

12:30 AM: 17 Stack Overflow tabs. Still broken. **0 marks.**

Three Problems, Three Solutions

Every reproducibility failure comes from losing control of one of three things:

Problem	Symptom	Solution
The Machine	"It works on my laptop but not yours"	Docker
The Math	"I get different results every run"	Seeds & determinism
The Memory	"Which of my 50 runs was the good one?"	Experiment tracking (TrackIO)

Today we fix all three — starting with the biggest headache.

The Machine

Docker: Ship the Whole Computer

"Works on My Machine" — The Most Famous Lie in CS

You shared your code AND `requirements.txt`. Your friend installs everything.

```
$ python train.py
OSError: libgomp.so.1: cannot open shared object file
```

What happened? Your code depends on a system library on your Mac that doesn't exist on their Windows laptop.

Virtual environments isolate **Python packages**. They **don't** isolate:

- Operating system (Mac vs Windows vs Linux)
- System libraries (C compilers, CUDA)
- Python version itself

We need to ship the entire computer, not just the package list.

The Shipping Container Analogy

Before shipping containers, every port loaded cargo differently — different cranes, different sizes, breakage everywhere. Then someone said: *"What if we put everything in one standard box?"*

Docker is that box for software.

You pack your code, Python, libraries, and OS into one standard container. It runs identically on any machine — your laptop, your friend's laptop, a cloud server.

Without Docker	With Docker
45 mins of `pip install` errors	`docker run spam-app` (2 min)
"Which sklearn version?"	Frozen inside the container
"Are you on Windows?"	Doesn't matter

That's literally why the logo is a whale carrying containers!

Why Docker? Four Scenarios

1. The Assignment Defense

You submit your ML assignment. TA runs it on Windows → crash.
With Docker: TA runs `docker run your-project` → guaranteed identical results.

2. The Dependency Minefield

Cool GitHub repo needs Python 3.7 + old PyTorch. Installing it breaks your other projects.
Docker: isolated bubble, no damage.

3. The Group Project Peacemaker

4 teammates: Windows, Mac Intel, Mac Silicon, Linux.
Docker gives everyone the exact same environment.

4. The Portfolio Builder

You built a Gradio demo. Cloud providers want a Docker image, not your laptop.

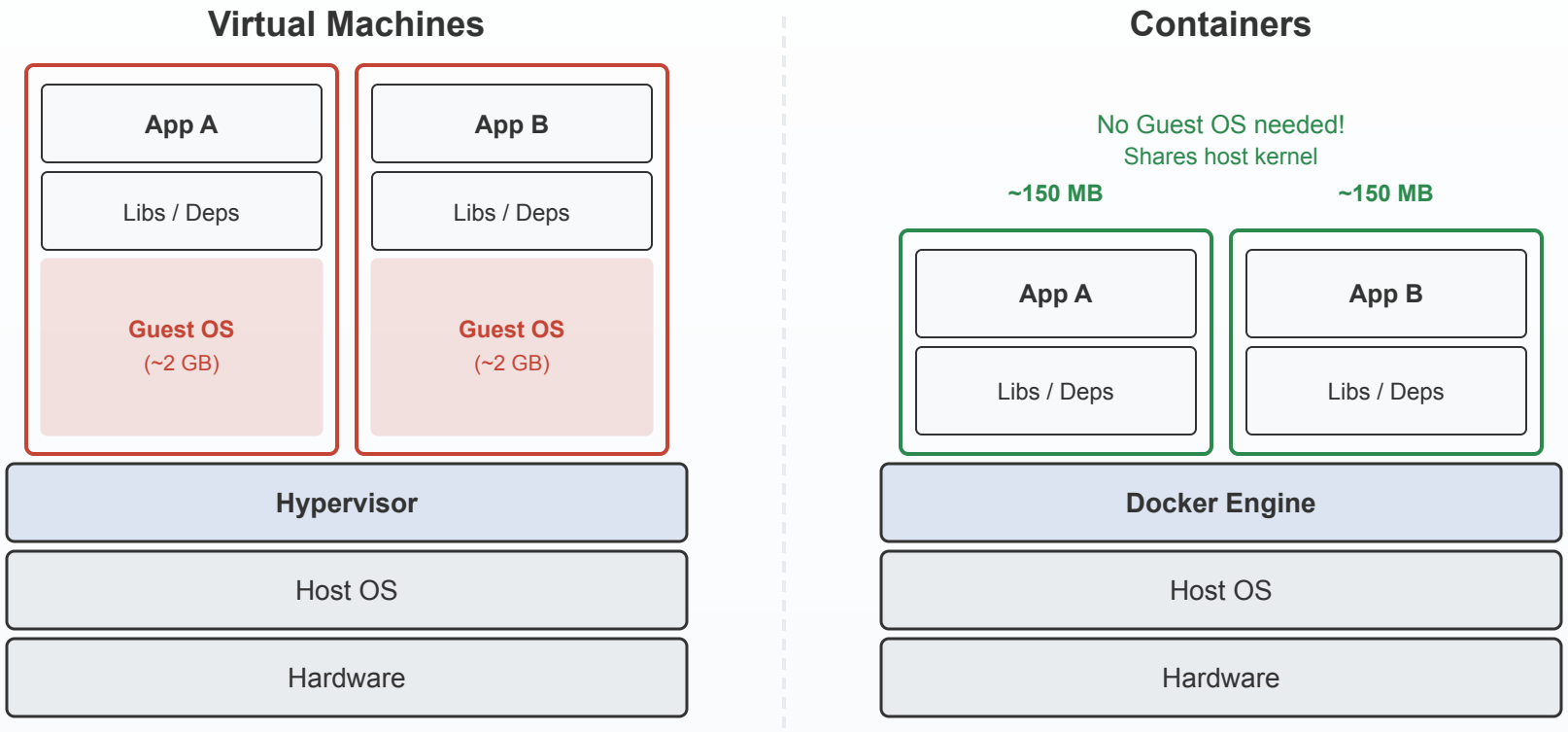
Six Concepts — That's All

Concept	What It Is	Analogy
Image	Blueprint — OS + libraries + code (read-only)	Recipe card
Container	Running instance of an image	Dish being cooked
Dockerfile	Instructions to build an image	The recipe
Docker Hub	Registry of pre-built images	GitHub for images
Volumes	Shared folder between container and laptop	USB drive
Ports	How you access container's web apps	Window into the kitchen

If Image is a **class**, Container is an **object** (instance).

Docker vs Virtual Machines

"Isn't Docker just a VM?"



VMs run a **full guest OS** (Windows inside Mac). Docker shares the host kernel — no guest OS needed.

Docker vs VMs: The Numbers

	VM	Docker Container
Size	GBs (full OS)	MBs (just your app)
Startup	Minutes	Seconds
Isolation	Complete (own kernel)	Process-level (shared kernel)
Use case	Run a different OS	Package an app with dependencies
RAM	Each VM needs GBs	Containers share host memory

Bottom line: VMs are heavy but fully isolated. Docker is lightweight and fast — perfect for shipping apps.

Docker Hub: The App Store for Images

`FROM python:3.10-slim` — where does this come from?

Docker Hub (hub.docker.com) — a public registry of pre-built images.

```
docker pull python:3.10-slim    # download an image
docker images                    # list images on your machine
```

Anyone can publish images. Official ones (Python, Ubuntu, nginx) are verified.

Docker Hub: Choosing the Right Image

Image	What's Inside	Size
`python:3.10`	Full Debian + Python + compilers	~1 GB
`python:3.10-slim`	Minimal Debian + Python	~150 MB
`python:3.10-alpine`	Alpine Linux + Python	~50 MB
`ubuntu:22.04`	Just Ubuntu, no Python	~77 MB

Always use `-slim` unless you need C compilers.

`-alpine` is smallest but can break packages that need `glibc`. Stick with `-slim`.

The Dockerfile: Line by Line

```
FROM python:3.10-slim              # 1. Start with Python + Linux

WORKDIR /app                       # 2. Create a workspace folder

COPY requirements.txt .            # 3. Bring the ingredient list
RUN pip install -r requirements.txt # 4. Install packages

COPY app.py spam_model.pkl ./      # 5. Bring our code + model

ENV GRADIO_SERVER_NAME="0.0.0.0"   # 6. Let the outside world connect

EXPOSE 7860                        # 7. Document the port

CMD ["python", "app.py"]           # 8. What runs when container starts
```

Line 6 is critical: Gradio defaults to `127.0.0.1` (container talks to itself only). `0.0.0.0` says "accept connections from outside."

Layer Caching: Why Second Build is Fast

Docker caches each step. If nothing changed, it reuses the cache:

```
$ docker build -t spam-app .
⇒ CACHED [1/5] FROM python:3.10-slim      # already downloaded
⇒ CACHED [2/5] WORKDIR /app               # nothing changed
⇒ CACHED [3/5] COPY requirements.txt .    # file unchanged
⇒ CACHED [4/5] RUN pip install ...        # packages cached!
⇒ [5/5] COPY app.py spam_model.pkl ./     # code changed → rebuilds
```

First build: 2 minutes (downloads everything).
Second build: 5 seconds (only re-copies your code).

Pro tip: `COPY requirements.txt` before `COPY . .` — if only your code changed, Docker skips the slow `pip install`!

Live Demos

Clone and follow along:

```
git clone https://github.com/nipunbatra/docker-ml-demos
```

Demo 1: Hello Docker

```
cd docker-ml-demos/1-hello
docker build -t hello-docker .
docker run hello-docker
```

What you see: Python version and OS are different from your laptop. It says "Linux" even on a Mac!

Try the interactive mode:

```
docker run -it hello-docker bash       # shell into the container
docker run -it hello-docker python     # Python REPL inside container
```

`-it` connects your keyboard + terminal. Without it, Docker runs headless.

See `WALKTHROUGH.md` in the repo for the full guided script.

Demo 2: Reproducible Dependencies

```
cd ../2-dependencies
docker build -t train-model .
docker run train-model          # always the same accuracy!
docker run train-model          # same again!
```

Pinned `scikit-learn==1.7.2` + `random_state=42` = identical results every time.

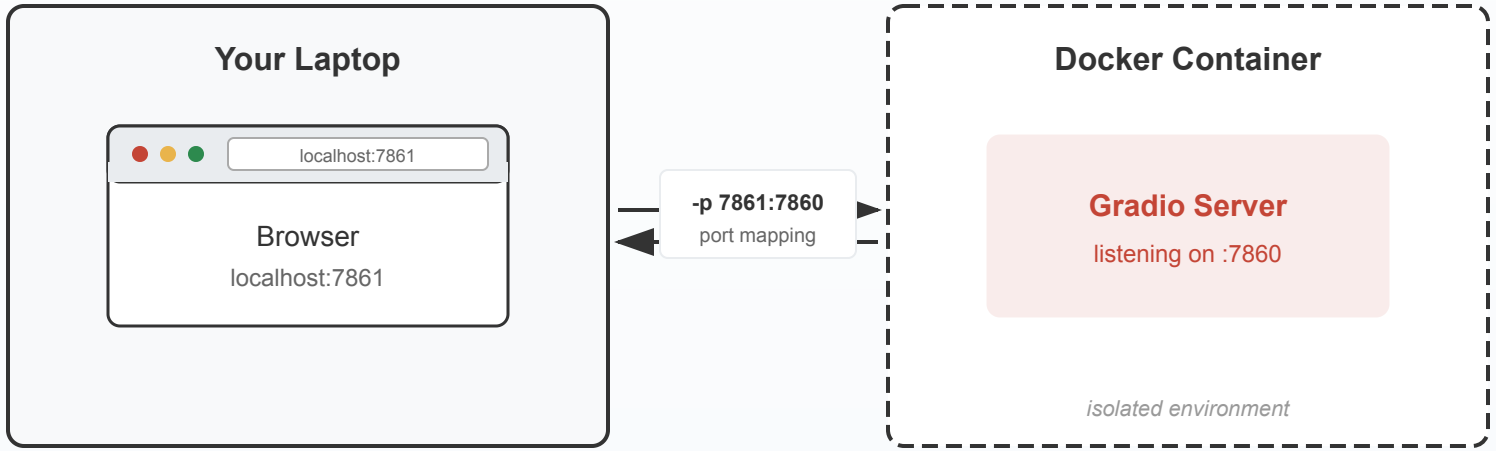
Compare with your laptop:

```
python -c "import sklearn; print(sklearn.__version__)"
# → probably different!
```

Demo 3: Web App + Port Mapping

```
cd ../3-web-app
docker build -t spam-app .
docker run -p 7861:7860 spam-app
```

Open http://localhost:7861 — a spam classifier is live!



`-p 7861:7860` = your laptop's port 7861 → container's port 7860.

Demo 4: Volumes — Containers Have Amnesia

```
cd ../4-volumes
docker build -t train-save .
docker run train-save                # model saved... inside container
ls outputs/                           # → No such file! Gone forever.

docker run -v $(pwd)/outputs:/app/outputs train-save
ls outputs/                           # → model.pkl, training_log.txt!
```

`-v` mounts a folder from your laptop into the container — like plugging in a USB drive.

Without `-v`, files created inside the container vanish when it stops. **By design.**

Demo 5: Environment Variables

```
cd ../5-environment
docker build -t env-demo .

# Default: RandomForest, 100 trees
docker run -p 7861:7860 env-demo

# Switch to SVM — no rebuild!
docker run -p 7861:7860 -e MODEL_TYPE=svm env-demo

# 500 trees — no rebuild!
docker run -p 7861:7860 -e N_ESTIMATORS=500 env-demo
```

Same image, three different behaviors. `-e` injects config at runtime.

In Python: `os.environ.get("MODEL_TYPE", "rf")`

Common Docker Errors

1. `docker: command not found`
→ Docker Desktop isn't running. Open the app first!

2. `port is already allocated`
→ Something else is using that port. Pick a different one: `-p 7862:7860`

3. `COPY failed: file not found`
→ The file isn't in the same folder as your Dockerfile. Check with `ls`.

4. Image is 3GB!
→ Use `FROM python:3.10-slim` instead of `python:3.10`

```
docker system prune   # clean up old images eating your disk
```

The 6 Docker Commands You Need

```
docker build -t my-app .              # Build image from Dockerfile
docker run my-app                      # Run (foreground)
docker run -d -p 7861:7860 my-app     # Run (background + port)
docker ps                              # List running containers
docker stop <id>                       # Stop a container
docker exec -it <id> bash             # Shell into running container
```

That's it. Docker solves the machine.

But even inside the same Docker container, you can still get different results every run...

The Math

Seeds & Determinism

"Yesterday 92%, Today 85%, Same Code?!"

Run your model training twice — exact same code:

```
X_train, X_test, y_train, y_test = train_test_split(X, y)
model = RandomForestClassifier()
model.fit(X_train, y_train)
print(model.score(X_test, y_test))
```

Run 1: 0.92 **Run 2:** 0.85 **Run 3:** 0.88

Which result do you report? The problem: **uncontrolled randomness.**

Sources of randomness in ML:

- **Train/test split** — which samples go where
- **Model initialization** — starting weights (neural nets)
- **Data shuffling** — order during training

The Minecraft Seed

In Minecraft, the world is randomly generated. But enter the **same seed** → get the **exact same world** every time.

Minecraft	Machine Learning
Seed number	`random_state=42`
Same seed → same world	Same seed → same split, same model
Share seed with friend → same map	Share seed with TA → same results

A random seed locks the universe into one specific timeline.

How to Lock the Universe

In sklearn: set `random_state` everywhere

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(n_estimators=100, random_state=42)

kf = KFold(n_splits=5, shuffle=True, random_state=42)
```

Global seed (for larger projects):

```
import random, numpy as np

def set_seed(seed=42):
    random.seed(seed)
    np.random.seed(seed)

set_seed(42)  # call once at the top of your script
```

Seeds: The Limits

Setting seeds makes YOUR code deterministic. But results can still differ across machines:

- **Different OS** → different BLAS/LAPACK math libraries under the hood
- **Different library versions** → sklearn 1.2 vs 1.4 may split data differently
- **GPU non-determinism** → CUDA operations are not always reproducible

That's why we need Docker (same OS + same libraries) **and** seeds (same randomness).

Seeds + Docker solve the math and the machine. But you still need to remember what you did...

The Memory

Experiment Tracking with TrackIO

The Spreadsheet Graveyard

You've been tuning hyperparameters all week:

```
run_lr001_depth5_est100.py     → 87.2%
run_lr01_depth10_est200.py     → 91.4%  ← wait, was this the one?
run_lr01_depth10_est200_v2.py  → 89.1%  ← or this?
run_FINAL.py                   → ???
run_FINAL_FINAL.py             → ???
run_FINAL_USE_THIS.py          → definitely not this
```

You had **the** best model last Tuesday. Now you can't find it.

Seeds made results reproducible. But you didn't record what you did.

TrackIO: Three Calls Is All You Need

TrackIO (by Hugging Face) — free, local-first experiment tracking.

```python
import trackio

# 1. Start a run
trackio.init(project="cs203-week08-demo", name="RandomForest",
             config={"model": "RF", "n_estimators": 100, "max_depth": 10})

# 2. Log metrics
trackio.log({"train_accuracy": 0.95, "test_accuracy": 0.845})

# 3. Finish
trackio.finish()
```

Everything stored locally in SQLite — no account, no cloud, no cost.

Demo: `python trackio_1_training_curves.py`

TrackIO: Training Curves

Log metrics at each step → dashboard shows curves:

```
trackio.init(project="cs203-week08-demo", name="gb-training",
             config={"model": "GradientBoosting", "lr": 0.1})

for n_est in range(10, 310, 10):
    gb = GradientBoostingClassifier(n_estimators=n_est, learning_rate=0.1,
                                    max_depth=3, random_state=42)
    gb.fit(X_train, y_train)
    trackio.log({
        "n_estimators": n_est,
        "train_accuracy": float(round(gb.score(X_train, y_train), 4)),
        "test_accuracy": float(round(gb.score(X_test, y_test), 4)),
    })

trackio.finish()
```

Watch accuracy climb and plateau — just like TensorBoard, but simpler.

TrackIO: Comparing Runs

Run the same model with 3 learning rates → dashboard **overlays** them:

```
for lr in [0.01, 0.1, 0.5]:
    trackio.init(project="cs203-week08-demo", name=f"lr-{lr}",
                 config={"learning_rate": lr})

    for n_est in range(10, 210, 10):
        gb = GradientBoostingClassifier(n_estimators=n_est, learning_rate=lr,
                                        random_state=42)
        gb.fit(X_train, y_train)
        trackio.log({
            "n_estimators": n_est,
            "train_accuracy": float(round(gb.score(X_train, y_train), 4)),
            "test_accuracy": float(round(gb.score(X_test, y_test), 4)),
        })

    trackio.finish()
```

Demo: `python trackio_3_compare_hyperparams.py`

TrackIO: Images & Tables

Log images to see what went wrong:

```
trackio.log({
    "misclassified": trackio.Image("errors.png",
        caption="All misclassified digits"),
})
```

Log tables for structured analysis:

```
trackio.log({"per_class_metrics": trackio.Table(
    dataframe=pd.DataFrame(table_data,
        columns=["Digit", "Precision", "Recall", "F1"]),
)})
```

Demo: `python trackio_2_misclassified.py` and `trackio_4_per_class_table.py`

TrackIO: Alerts for Overfitting

Get notified when something goes wrong:

```
for depth in range(1, 30):
    dt = DecisionTreeClassifier(max_depth=depth, random_state=42)
    dt.fit(X_train, y_train)
    gap = dt.score(X_train, y_train) - dt.score(X_test, y_test)

    trackio.log({"max_depth": depth, "overfit_gap": gap})

    if gap > 0.08:
        trackio.alert(title="Overfitting detected!",
                      text=f"Gap={gap:.1%} at depth={depth}",
                      level=trackio.AlertLevel.ERROR)
```

Demo: `python trackio_5_overfitting_alert.py`

TrackIO: The Dashboard

```
pip install trackio
trackio show --project cs203-week08-demo
```

Tab	What You See
Metrics	Training curves, overlaid runs
Media & Tables	Images, per-class tables
Runs	All configs and final metrics
System Metrics	GPU/CPU usage (auto on Apple Silicon)

The Complete Reproducibility Stack

Layer	Tool	What It Controls
The Machine	Docker	OS + system libs + everything
The Packages	`venv` + `requirements.txt`	Library versions
The Math	`random_state=42`	Algorithmic randomness
The Memory	TrackIO	What you tried & what worked

```
Docker + Venv + Seeds + TrackIO = Time Capsule
```

You don't always need all layers. For course projects: seeds + TrackIO + venv is enough. Add Docker when shipping to production or sharing across OS.

Key Takeaways

1. **Docker ships the machine** — OS + libraries + Python + your code, frozen together
2. **Set seeds everywhere** — `random_state=42` in every sklearn call
3. **Track experiments** — `trackio.init()` / `log()` / `finish()` — never lose a good run
4. **Pin dependencies** — `pip freeze > requirements.txt` with exact versions
5. **Volumes for persistence** — containers have amnesia; use `-v` to save data
6. **Reproducibility is a gift to your future self** — if no one else can run it, it might as well not exist